12.367

EE25BTECH11042 - Nipun Dasari

Question:
The eigen-values of a real symmetric matrix are always

a) positive
b) imaginary
c) real
d) complex conjugate pairs

Solution:

The correct option is (c), the eigenvalues of a real symmetric matrix are always **real**.

Proof without Hermitian Conjugate:

Let $\mathbf{A}$ be a real symmetric matrix. This means all its entries are real numbers and $\mathbf{A} = \mathbf{A}^\top$.

Let λ be an eigenvalue of $\mathbf{A}$ and $\mathbf{v}$ be its corresponding eigenvector. Note that λ and the components of $\mathbf{v}$ could be complex. The eigenvalue equation is:

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v} \tag{4.1}$$

If we take the complex conjugate of this entire equation, we get:

$$(\mathbf{A}\mathbf{v})^c = (\lambda\mathbf{v})^c \tag{4.2}$$

$$\mathbf{A}^c\mathbf{v}^c = \lambda^c\mathbf{v}^c \tag{4.3}$$

Since $\mathbf{A}$ is a **real** matrix, its complex conjugate is itself ($\mathbf{A}^c = \mathbf{A}$). So, the equation becomes:

$$\mathbf{A}\mathbf{v}^c = \lambda^c\mathbf{v}^c \tag{4.4}$$

Now, left-multiply the original eigenvalue equation by $(\mathbf{v}^c)^\top$:

$$(\mathbf{v}^c)^\top\mathbf{A}\mathbf{v} = (\mathbf{v}^c)^\top(\lambda\mathbf{v}) = \lambda((\mathbf{v}^c)^\top\mathbf{v}) \tag{4.5}$$

Next, take the transpose of both sides of equation (4.4):

$$(\mathbf{A}\mathbf{v}^c)^\top = (\lambda^c\mathbf{v}^c)^\top \tag{4.6}$$

$$(\mathbf{v}^c)^\top\mathbf{A}^\top = \lambda^c(\mathbf{v}^c)^\top \tag{4.7}$$

Since $\mathbf{A}$ is a **symmetric** matrix, $\mathbf{A}^\top = \mathbf{A}$. The equation is:

$$(\mathbf{v}^c)^\top\mathbf{A} = \lambda^c(\mathbf{v}^c)^\top \tag{4.8}$$

Now, right-multiply this equation by $\mathbf{v}$:

$$(\mathbf{v}^c)^\top\mathbf{A}\mathbf{v} = \lambda^c((\mathbf{v}^c)^\top\mathbf{v}) \tag{4.9}$$

By comparing equations (4.5) and (4.9), we see that:

$$\lambda((\mathbf{v}^c)^\top \mathbf{v}) = \lambda^c((\mathbf{v}^c)^\top \mathbf{v}) \tag{4.10}$$

$$(\lambda - \lambda^c)((\mathbf{v}^c)^\top \mathbf{v}) = 0 \tag{4.11}$$

The term $(\mathbf{v}^c)^\top \mathbf{v}$ is the dot product of the vector $\mathbf{v}^c$ with $\mathbf{v}$, which is the squared norm of $\mathbf{v}$.

$$(\mathbf{v}^c)^\top \mathbf{v} = \sum_i {v_i}^c v_i = \sum_i |v_i|^2 = ||\mathbf{v}||^2 \tag{4.12}$$

Since $\mathbf{v}$ is an eigenvector, it cannot be the zero vector, so its norm $||\mathbf{v}||^2$ must be a positive real number.

Given that $||\mathbf{v}||^2 \neq 0$, the only way for the equation $(\lambda - \lambda^c)\,||\mathbf{v}||^2 = 0$ to hold is if:

$$\lambda - \lambda^c = 0 \implies \lambda = \lambda^c \tag{4.13}$$

A complex number is equal to its own conjugate only if its imaginary part is zero. Therefore, λ must be a real number.